

A Century of Natural Disasters: More Events, Fewer Deaths, Bigger Bills

Visual Exploration of the EM-DAT Emergency
Events Database, 1900–2022

Programming Assignment 1 (A1)

Visual Exploration Using Visual Analytics Tools

- Dataset allotted** #7 — Natural Disasters Emergency Events Database (EM-DAT based, 1900–2022)
- Team size** 3 members
- Submission** 13 September 2026

Team and task allocation

Member	Roll number	Task set
Lakshya Jain	BT2024044	A — <i>When?</i> The temporal record (Figures 3–7), plus data preprocessing
Arnav Jain	BT2024233	B — <i>Where?</i> The geography of exposure (Figures 8–12)
Kalpit Phogat	BT2024093	C — <i>What?</i> The impact signature of hazards (Figures 13–17)

Note on AI use. Each team member submits a signed AI declaration separately, on the form circulated by the course instructor. The verification harness described in Section 11 documents how every number in this report was independently re-derived from the raw data.

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1 Introduction and Guiding Question

Public conversation about natural disasters usually runs on a single intuition: that the world is being hit harder every year. The EM-DAT Emergency Events Database, which has catalogued disaster events country by country since 1900, is the standard evidence base for testing that intuition. This assignment uses it to answer one question through visualization.

Guiding question. Over 1900–2022, has the burden of natural disasters actually grown, and has the world become better or worse at surviving it?

The question is deliberately compound, because a disaster has no single cost. An event can be *counted*, it can *kill*, it can *displace*, and it can *destroy property*, and those four measures need not move together. The question therefore decomposes into the three sub-questions of Table 1, each owned end to end by one team member so that no task is fragmented across the team. Figure 1 shows that decomposition and how the three strands converge on a single data story.

Table 1. The guiding question decomposed into three independently owned sub-questions. Ownership is by whole task set, never by individual chart.

	Sub-question	Owner	Figures
A	When? How has the recorded burden changed over time — and how much of that change is real rather than an artefact of record-keeping?	Member 1	3–7
B	Where? Where does the burden fall, and does the country that records the most disasters also suffer the most harm?	Member 2	8–12
C	What? Which hazards kill, which displace, and which cost money — and are they the same hazards?	Member 3	13–17

Each sub-question is answered by a *sequence* of visualization tasks rather than by a single chart, and the three sequences are designed to converge on one data story, set out in Section 8.

1.1 Visualization tasks

Tasks are labelled using the descriptive verbs of the design space of Schulz et al. [1]: *overview*, *trend*, *locate*, *rank*, *compare*, *relate*, *derive*, *correlate*, *identify*, *validate*. Table 2 lists the fifteen tasks. Each figure in Sections 4–6 is introduced by the task it implements, the rationale for the chart form chosen, and the inference drawn from it.

Table 2. The fifteen visualization tasks, their Schulz et al. verbs, and the figure that implements each one.

Task	Description	Verb(s)	Figure
A1.1	Characterise the composition of the record	Overview, Summarize	3
A1.2	Trace recorded events through time	Trend	4
A1.3	Test whether the trend is real or a reporting artefact	Identify, Validate	5
A1.4	Contrast total mortality against mortality per event	Compare, Derive	6
A1.5	Test whether the mortality decline is broad-based	Validate, Derive	7
B1.1	Locate disaster occurrence on the globe	Locate, Overview	8
B1.2	Locate disaster mortality on the globe	Locate, Compare	9
B1.3	Rank countries on four separate burdens	Rank, Compare	10
B1.4	Quantify how concentrated each burden is	Derive, Compare	11
B1.5	Derive lethality with exposure held roughly constant	Derive, Rank	12
C1.1	Compare each hazard’s share of four burdens	Compare, Summarize	13
C1.2	Relate a hazard’s deadliness to its cost	Relate, Locate	14
C1.3	Trace real economic damage through time	Trend	15
C1.4	Compare lethality change across hazards and decades	Compare, Trend	16
C1.5	Correlate the four burdens on one indexed scale	Correlate, Trend	17

1.2 Tasks considered and rejected

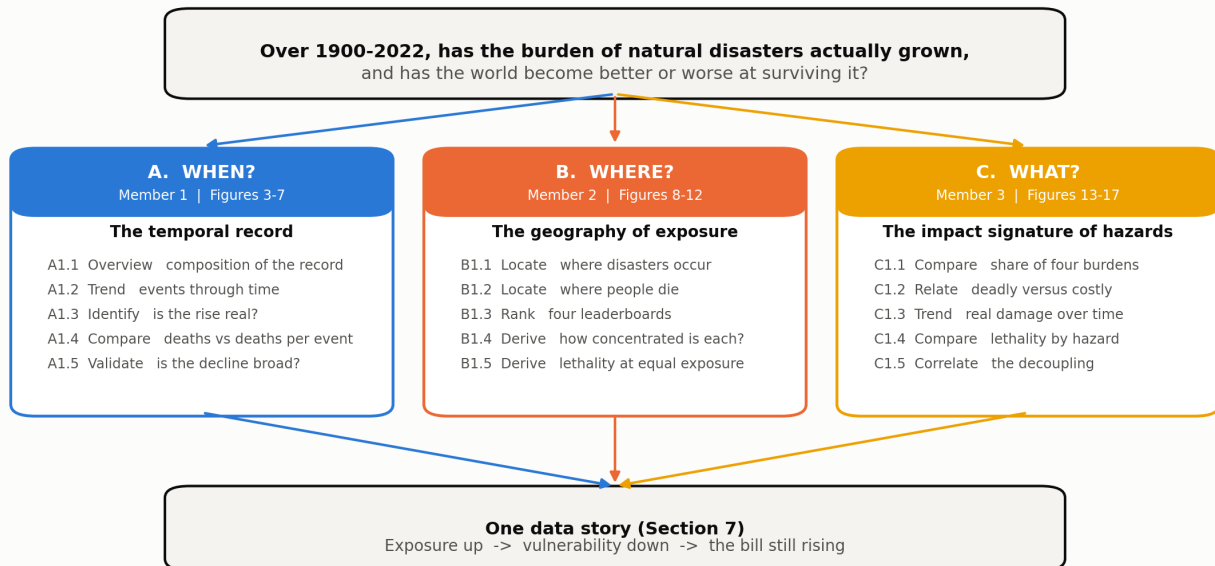
The brief asks for the tasks we *tried* as well as those we eventually implemented. Six candidate tasks were prototyped and dropped, each for a reason that is itself a finding about the dataset.

1. **Deaths per million population, by country.** The natural way to compare countries fairly. Dropped because the file carries no population column; supplying one would mean joining an external source with a different vintage and country list, which we judged out of scope for a single-dataset assignment. Task B1.5 (Figure 12) is the substitute: it holds *exposure* roughly constant instead of population, by restricting to the 20 countries with the most recorded events.
2. **Damage as a share of GDP.** Same problem as population, and the same resolution: damage is reported per event and per decade in constant dollars instead.
3. **Epidemics and pandemics.** Attempted, then abandoned on inspection: the Biological subgroup in this file contains only 92 insect infestations and one animal accident. There are no epidemic rows at all, so no honest task can be built on them.
4. **Sub-type analysis** (25 sub-types, e.g. riverine flood versus flash flood). Prototyped and dropped: 20.4% of rows carry no sub-type, and the missingness is concentrated in the early decades. A sub-type time series would therefore have shown a spurious “rise of flash floods” that is really the arrival of finer classification. The analysis stays at the 13-type level, where coverage is complete.
5. **A country $\times$ hazard-type matrix.** Built as a clustered heatmap, then dropped: with 225 countries it is unreadable at report scale and violates the brief’s requirement for plots a layman can read. Its one useful finding is stated in text instead — of the 132 countries with at least 20 recorded events, flood is the most frequent hazard in 82 and storm in 39.
6. **A choropleth of deaths per event.** Dropped because countries with one or two recorded events produce extreme, meaningless ratios that dominate the colour scale. Figure 12 shows the same measure as a ranked bar chart over a filtered, comparable set of countries.

Two further presentation ideas were rejected on principle rather than on data: a dual-axis chart pairing deaths with damage (see Section 3), and an animated time slider, which cannot survive into a static PDF and is instead part of the Tableau demo.

One question, three independently owned task sets, fifteen charts

Each member owns a sub-question end to end - its tasks, its figures, its inferences and its section of the report and of the video - so no task is split across the team. Task verbs follow the design space of Schulz et al.



Source: EM-DAT Emergency Events Database country profiles (CRED / UCLouvain), 1900-2022 snapshot of 2023-04-06.

Figure 1. Task decomposition. One guiding question, three independently owned task sets, fifteen charts, one data story. Ownership is by whole sub-question so that each member's contribution is separable and equal, as the assignment brief requires.

2 Dataset Description

2.1 Source and provenance

The file used is the **country-profiles** export of EM-DAT, the Emergency Events Database maintained by the Centre for Research on the Epidemiology of Disasters (CRED) at UCLouvain, Brussels [2], distributed on Kaggle as *Natural Disasters Emergency Events Database* [3]. The single CSV in that dataset is

`_EmergencyEventsDatabase-CountryProfiles_emdat-country-profiles_2023_04_06.csv`

a snapshot taken on **6 April 2023**. It is semicolon-delimited and uses a comma as the decimal mark, so it must be read with `sep=';', decimal=','`; read with defaults, every numeric column arrives as text.

A row is **not** one disaster. A row is one (*year, country, disaster type, disaster subtype*) combination, with the number of events in that combination held in **Total Events**. There are **10,431 rows** describing **15,015 recorded events**.

2.2 Schema

Table 3 lists the thirteen columns, the variable type of each and how often it is missing. The dataset spans all four variable types the brief asks for: temporal (**Year**), categorical (five nested hazard fields), numerical (four impact measures) and geospatial (**Country / ISO**).

Table 3. Schema of the raw file, with the variable type of each column and the share of rows in which it is missing.

Column	Type	Description	Missing
Year	Temporal (ordinal)	1900–2023	0%
Country	Categorical (nominal)	225 distinct countries and territories	0%
ISO	Categorical (nominal)	ISO-3166 alpha-3 code; the join key for the maps	0%
Disaster Group	Categorical	Always Natural ; no technological disasters	0%
Disaster Subroup	Categorical	5 values: Hydrological, Meteorological, Geo-physical, Climatological, Biological (<i>the column name is misspelled in the source file</i>)	0%
Disaster Type	Categorical	13 values, e.g. Flood, Storm, Earthquake, Drought	0%
Disaster Subtype	Categorical	25 values, e.g. Riverine flood, Tropical cyclone, Ground movement	20.4%
Total Events	Numerical (count)	Events in that year/country/type/subtype cell; 1–20	0%
Total Affected	Numerical (count)	People injured, made homeless, or requiring assistance	27.3%
Total Deaths	Numerical (count)	Deaths and missing presumed dead	29.3%
Total Damage (USD, original)	Numerical (currency)	Damage in the prices of the year of the event	63.2%
Total Damage (USD, adjusted)	Numerical (currency)	Damage in constant 2022 US dollars	63.2%
CPI	Numerical (index)	Price index used for the adjustment; 2.85 (1900) to 100.00 (2022)	0.5%

Missing percentages are quoted for the raw 10,431-row file. In the 10,380-row analysis frame the CPI column is complete, because the 51 rows that lacked it are exactly the 51 partial-year 2023 rows that are dropped.

We verified the relationship between the two damage columns rather than assuming it. Across all rows where both are present,

$$\text{Total Damage (adjusted)} = \text{Total Damage (original)} \times \frac{100}{\text{CPI}}$$

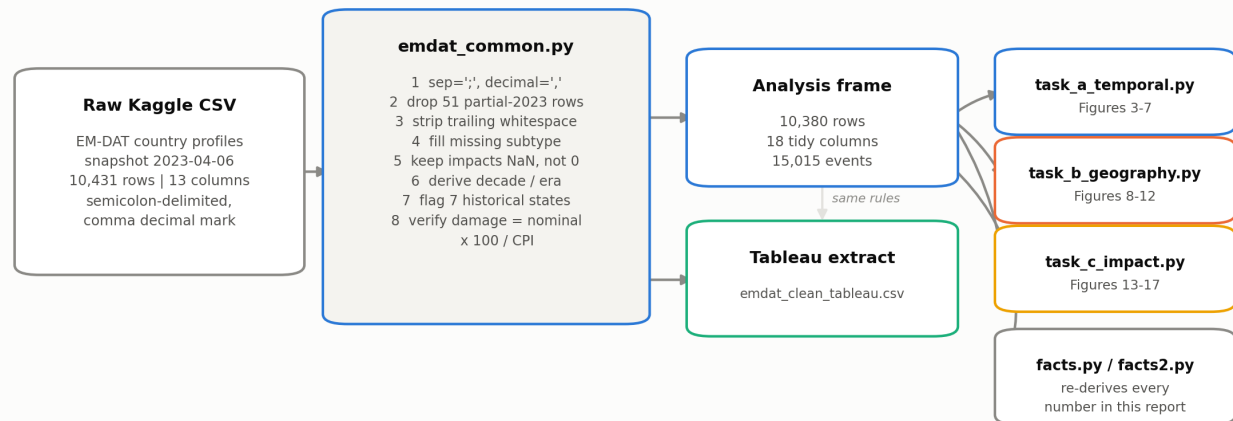
holds to within 2.5×10^{-4} , and $\text{CPI} = 100$ exactly in 2022. **The adjusted column is therefore constant 2022 US dollars, and it is the only damage measure used in this report.** Comparing nominal dollars across a 123-year span would have made the trend in Figure 15 almost entirely an artefact of inflation: the nominal total is \$4.19 trillion against a real total of \$6.70 trillion.

2.3 Preprocessing

All cleaning is performed in one shared module, `code/emdat_common.py`, so that the three task sets are guaranteed to be working from an identical frame. Figure 2 shows the pipeline; Table 4 lists the rules and the reason for each.

One cleaning module feeds every chart in this report

The three task sets never clean the data themselves, so all fifteen charts are guaranteed to describe an identical frame. The verification harness reads that same frame and re-derives every number quoted in this report.



Boxes are files in code/. The row counts shown are printed by the module itself on every run.

Figure 2. The processing pipeline. A single cleaning module produces the analysis frame and the Tableau extract; the three task-set scripts and the verification harness all read that same frame, so no chart in this report can be describing a differently-cleaned dataset.

Table 4. Preprocessing rules applied in `emdat_common.py`, their effect on the data, and the reason each is necessary.

#	Rule	Effect	Why
1	Read with <code>sep=';', decimal=','</code>	Numeric columns parse correctly	European CSV conventions; otherwise every measure is a string
2	Drop rows dated 2023	−51 rows → 10,380 rows	The snapshot is dated 6 April 2023, so 2023 is a fractional year that would fake a collapse in every trend. All 51 rows also have a null CPI, confirming they are incomplete
3	Strip whitespace from all label columns	13 hazard types, not 14	The source carries a trailing space on some Disaster Type values (" Extreme temperature "), silently splitting one category in two
4	Fill missing Disaster Subtype with "<Type> (unspecified)"	2,133 rows retained	EM-DAT records only a coarse type for older events; dropping them would bias the record toward recent years
5	Leave impact columns as NaN, never zero	Per-event severity stays honest	Zero-filling would treat “not reported” as “nobody died”, understating historical severity
6	Derive decade and a 1900–1969 / 1970–2022 era flag	Used by Figures 5, 6, 7, 15, 16	Decadal aggregation suppresses single-year noise
7	Flag 7 historical states no longer in existence	Soviet Union, Czechoslovakia, Yugoslavia, Germany Dem Rep, Serbia Montenegro, Yemen Arab Rep, Yemen P Dem Rep	These appear in the country rankings (Figure 10) and must be read as historical entities, not present-day countries
8	Emit the cleaned Tableau extract <code>data/emdat_clean_tableau.csv</code>	10,380 rows, 18 tidy columns	A flat, Tableau-ready extract needing no further cleaning

2.4 Known limitations of the source

These constrain what can honestly be claimed, and are revisited in Section 10.

1. **Reporting bias.** 90.6% of all recorded events fall in 1970–2022, a span covering only 43.1% of the 123 years. Event counts before about 1970 measure historical record-keeping at least as much as they measure hazard.
2. **Damage under-reporting.** Only 36.9% of records carry a damage figure, so every damage total here is a lower bound.
3. **No population denominator.** The file has no population column, so “deaths per event” is used as the severity measure rather than deaths per capita. A large country will record more events than a small one for reasons of area and population alone.
4. **No epidemics.** The **Biological** sub-group in this file contains only insect infestations (92 rows) and one animal accident. There are no epidemic or pandemic rows, so COVID-19 and comparable events are entirely absent.
5. **Aggregated rows.** Because a row can hold up to 20 events, per-event figures are averages over a cell, not properties of individual disasters.

3 Method and Design Conventions

Every figure in this report follows one set of rules, so that a reader who learns to read the first chart can read all fifteen.

- **Colour has one job per chart.** Categorical palettes encode identity (hazard type, sub-group) with a fixed slot order that never changes between figures — *Hydrological* is the same blue in Figure 4 and Figure 15. Sequential single-hue blue ramps encode magnitude (Figures 8, 9, 16). No rainbow scales are used.
- **Colour-vision safety was computed, not judged.** The five-colour categorical palette was run through a colour-vision-deficiency separation check before use; the worst adjacent pair scores $\Delta E = 9.1$ under protanopia against a target of ≥ 8 , and every value shown on a light-coloured fill carries a visible text label, so colour is never the only channel.
- **No dual-axis charts.** Where two measures of different scale must be compared, they are drawn as two stacked panels sharing an x -axis (Figure 6) or indexed to a common base (Figure 17). A twin y -axis lets the author choose the crossing point and is therefore avoided.
- **Log scales are labelled as such** and are used only where the data span three or more orders of magnitude.
- **Every chart states its own finding in its title.** The chart title is the inference; the chart subtitle carries the caveat. The L^AT_EX caption below each figure gives the rationale for the chart form.
- **Part-decade warning.** The 2020s bucket holds only 2020–2022 and is annotated wherever it appears.

4 Task Set A — *When?* The Temporal Record

Owner: Lakshya Jain · Figures: 3–7 · Script: `code/task_a_temporal.py`

Task A1.1 — Overview: what is in the record?

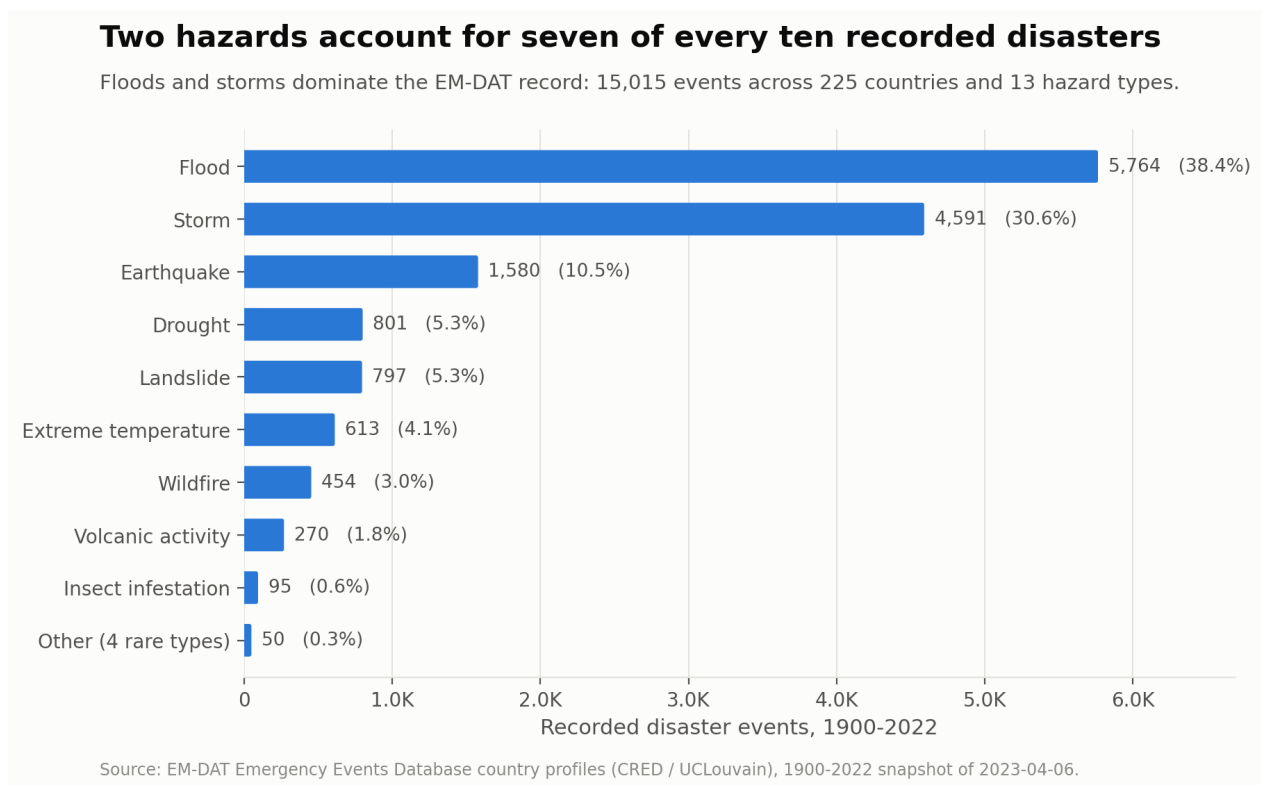


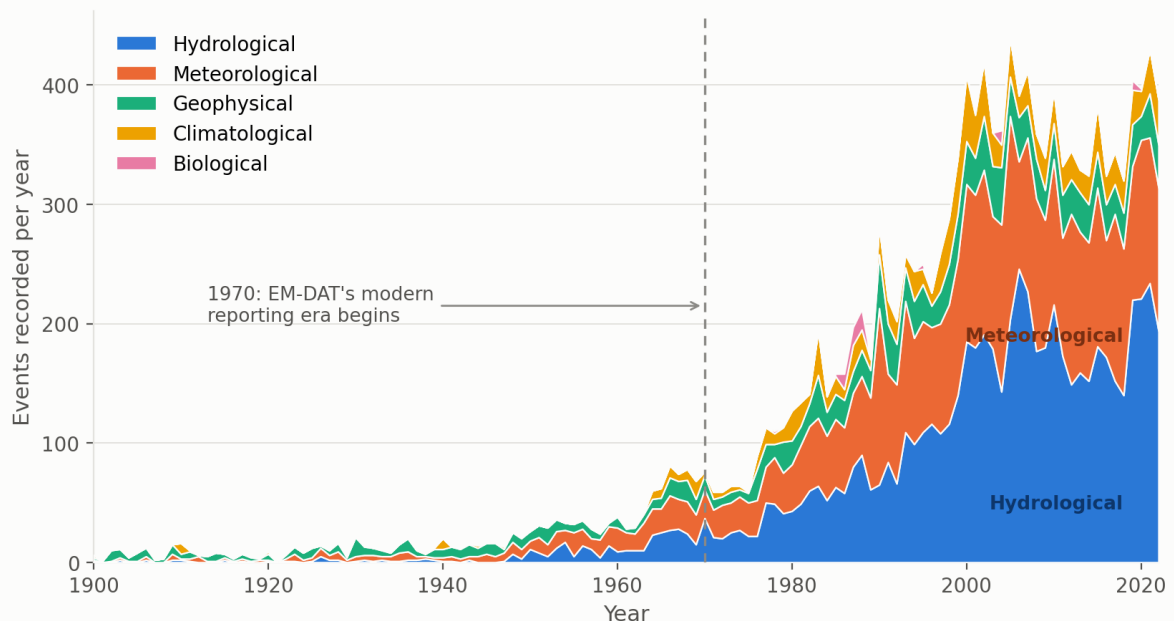
Figure 3. Recorded disaster events by hazard type, 1900–2022. A horizontal bar chart is used because the category labels are long words that read naturally on the *y*-axis, and because rank is the question being asked. The four rarest types (mass movement, glacial lake outburst, fog, animal accident — 50 events combined) are folded into “Other” rather than being given their own colours.

Inference. The record is dominated by water and weather. Floods (5,764 events, 38.4%) and storms (4,591, 30.6%) together account for **69% of all recorded events**. Earthquakes are a distant third at 10.5%. Drought — which Task Set C will show to be by far the deadliest hazard — accounts for only 5.3% of events. The first finding of the report is therefore that *frequency and harm are different questions*, and the rest of the analysis follows that split.

Task A1.2 — Trend: when were these events recorded?

The record explodes after 1970 - mostly floods and storms

91% of all events in the file are dated 1970–2022, a period covering only 43% of the 123 years.



Source: EM-DAT Emergency Events Database country profiles (CRED / UCLouvain), 1900–2022 snapshot of 2023-04-06.

Figure 4. Events recorded per year, stacked by disaster sub-group. A stacked area chart is used because the question is about both the total and its composition over a continuous time axis. The two largest bands are labelled directly on the chart in addition to the legend, so identity never depends on colour alone.

Inference. The curve is close to flat until roughly 1960 and then rises steeply, peaking at 440 events in 2005. The growth is overwhelmingly hydrological and meteorological — floods and storms — which is consistent with a climate-driven increase, but the shape of this curve is also exactly what a step-change in data collection would look like. **91% of all events in the file are dated 1970–2022, a span covering only 43% of the years.** That is a suspicious concentration, and Figure 5 tests it directly.

Task A1.3 — Identify: is the rise real, or is it record-keeping?

The count of disasters is partly a count of reporting

Volume grew 52x from the 1900s to the 2000s, but the share of records carrying an economic-damage figure peaks at just 50% - so every damage total in this report is a lower bound.

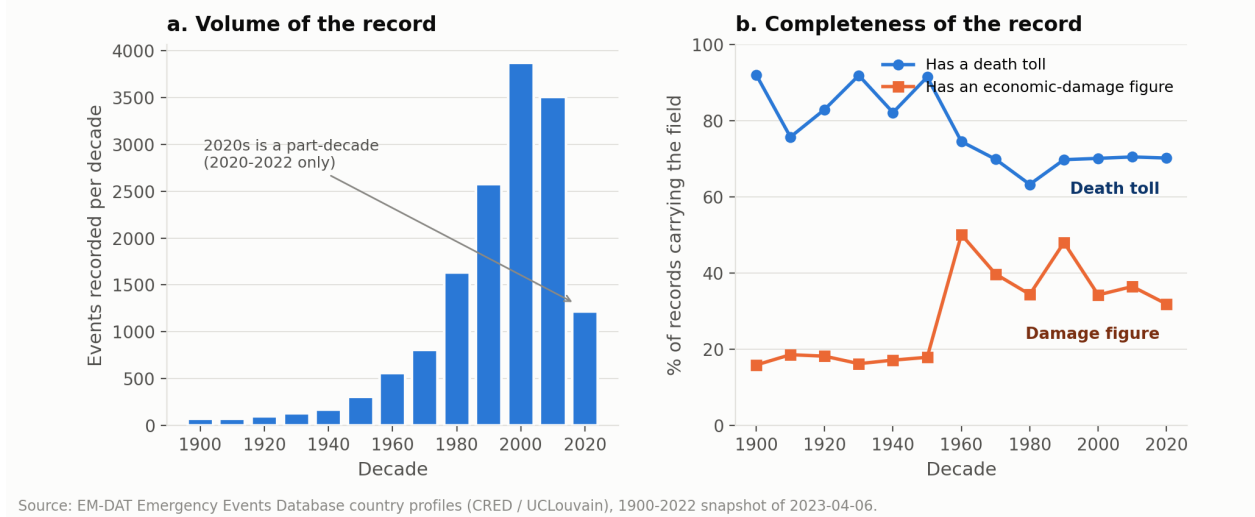


Figure 5. Volume and completeness of the record. Left: events recorded per decade. Right: the share of records that actually carry a death toll and a damage figure, by decade. Two panels rather than one chart, because volume and completeness are different measures; putting them on a shared *y*-axis would be meaningless.

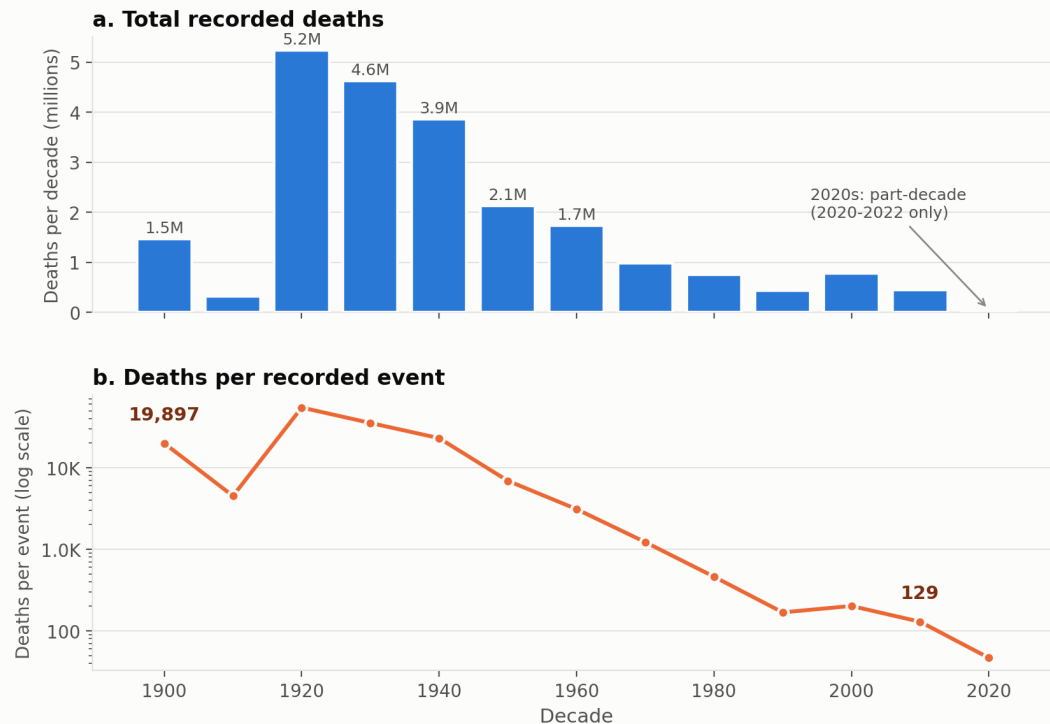
Rationale. This is the diagnostic figure of the report. Before drawing any conclusion from a rising count, we ask whether the instrument changed.

Inference. Recorded volume grew **52-fold** between the 1900s (74 events) and the 2000s (3,872 events) — a rate no plausible physical change in hazard frequency could produce on its own. Meanwhile, the share of records carrying an economic-damage figure peaks at just 50% in any decade, and sits between 32% and 48% across the entire post-1970 period. The honest reading is that the event count in Figure 4 is **a joint measurement of hazard and of reporting effort, and cannot be separated into the two**. Every subsequent figure in this report therefore prefers *rates* (deaths per event, share of total, damage per event) over raw counts, because a rate is far less sensitive to how many events were catalogued.

Task A1.4 — Compare: total mortality against mortality per event

Disasters became far more survivable even as they became far more numerous

Deaths per recorded event fell 99.4%, from ~19,900 in the 1900s to ~129 in the 2010s. The two measures use different scales, so they are drawn as two panels rather than on a twin axis.



Source: EM-DAT Emergency Events Database country profiles (CRED / UCLouvain), 1900-2022 snapshot of 2023-04-06.

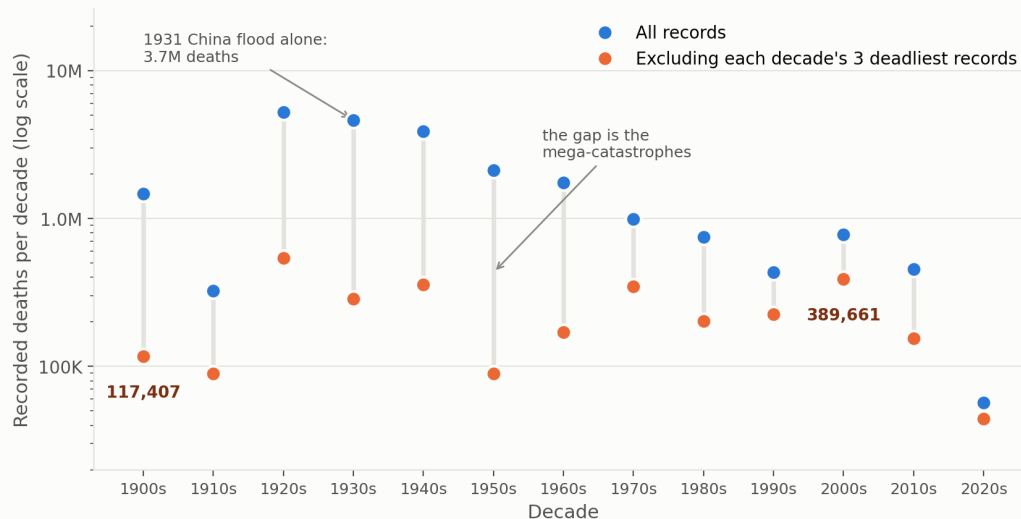
Figure 6. Total recorded deaths against deaths per recorded event. Panel (a) totals deaths by decade; panel (b) divides by the number of events, on a log scale. The two measures differ by five orders of magnitude, so they are drawn as two panels sharing one *x*-axis rather than on a twin *y*-axis.

Inference. The two panels point in opposite directions, and that contradiction is the centre of the data story. Total deaths peaked in the **1920s at 5.24 million** and have fallen to 0.45 million in the 2010s. Deaths per recorded event fell from **≈19,900 in the 1900s to ≈129 in the 2010s — a 99.4% decline**. Whatever has happened to the frequency of disasters, an individual recorded disaster today kills roughly **one 150th** of what one did a century ago. This is the single strongest signal in the dataset.

Task A1.5 — Validate: is the mortality decline broad-based?

The fall in deaths is the disappearance of mega-catastrophes, not a broad decline

Strip each decade's three deadliest records and the residual death toll shows no downward trend at all - the 2000s sit above the 1900s. A dot plot is used rather than bars because bar length is not proportional to value on a logarithmic axis.



Source: EM-DAT Emergency Events Database country profiles (CRED / UCLouvain), 1900-2022 snapshot of 2023-04-06.

Figure 7. Deaths per decade, with and without each decade's three deadliest records. A paired dot plot: each decade has one dot per treatment and a connector whose length is the contribution of the three deadliest records. A log scale is required because the values span three orders of magnitude — and that is precisely why dots are used rather than bars, since bar *length* is not proportional to value once the axis is logarithmic, whereas dot *position* still is. This figure is a deliberate attempt to break the result of Figure 6.

Rationale. A 99.4% fall is a large claim. Because EM-DAT's early decades are dominated by a handful of enormous famines and floods, we test whether the decline survives their removal.

Inference. It does not, in the form usually stated. Once each decade's three deadliest records are removed, the residual death toll shows **no downward trend whatsoever**: the 2000s (390,000) sit *above* the 1900s (117,000). The century-scale collapse in total mortality is almost entirely **the disappearance of mega-catastrophes** — single events such as the 1931 China flood (3.7 million deaths) and the 1928 China drought (3.0 million) — rather than a broad reduction in ordinary disaster deaths. This is a genuinely different conclusion from the headline one, and it is the kind of finding that only emerges from deliberately trying to break your own result.

5 Task Set B — *Where?* The Geography of Exposure

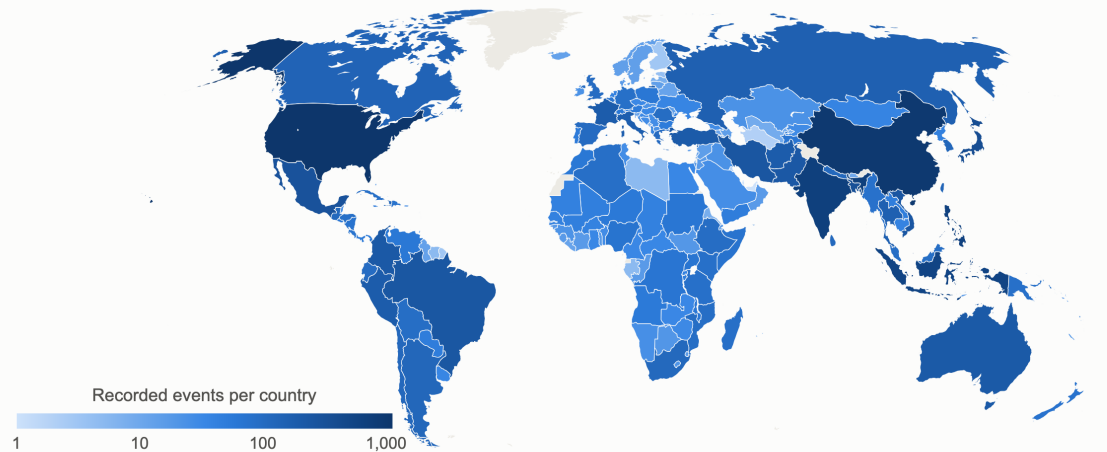
Owner: Arnav Jain · Figures: 8–12 · Script: code/task_b_geography.py

Task B1.1 — Locate: where are disasters recorded?

Almost every country on Earth appears in the disaster record

Total recorded disaster events per country, 1900-2022.

225 countries appear; the United States (1,119) and China (984) lead.



Source: EM-DAT Emergency Events Database country profiles (CRED / UCLouvain), 1900-2022 snapshot of 2023-04-06. Grey = no recorded event. Colour scale is logarithmic.

Figure 8. Total recorded disaster events per country, 1900–2022. A choropleth answers “where” directly. The colour scale is a single-hue sequential blue ramp on a *logarithmic* scale — country totals span five orders of magnitude, and a linear ramp would render every country except the United States and China identically pale.

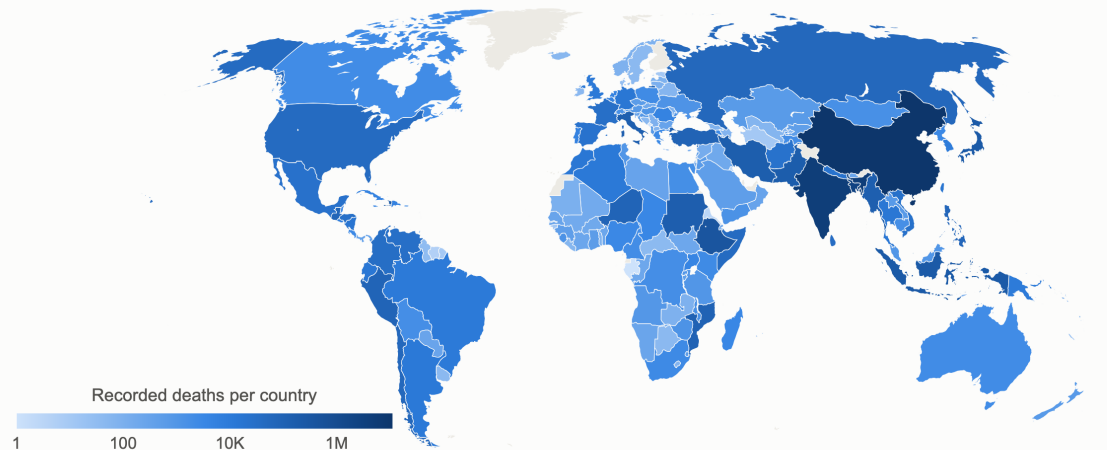
Inference. Disaster occurrence is close to universal: all **225** countries and territories in the file appear at least once. The darkest countries are the large, populous, well-documented ones — the United States (1,119 events), China (984), India (688), the Philippines (659) and Indonesia (570). Note that this map is as much a map of *area, population and reporting capacity* as of hazard.

Task B1.2 — Locate: where do people actually die?

But death is concentrated in a handful of countries

Total recorded disaster deaths per country, 1900-2022.

China and India alone account for more than half of all 22.9 million recorded deaths.



Source: EM-DAT Emergency Events Database country profiles (CRED / UCLouvain), 1900-2022 snapshot of 2023-04-06. Grey = no recorded event. Colour scale is logarithmic.

Figure 9. Total recorded disaster deaths per country, 1900–2022. Drawn on the same projection and the same style of logarithmic colour scale as Figure 8, so the two maps can be read directly against each other. Holding the encoding constant is what makes the comparison legible.

Inference. The map changes character completely. Where Figure 8 is broadly shaded, Figure 9 is dominated by two countries. **China (10.96 million) and India (4.59 million) alone account for 68% of all 22.86 million recorded deaths;** Bangladesh adds a further 2.59 million. The United States, which leads the world in *recorded events*, ranks **23rd** in deaths with 44,246. Occurrence and mortality are not the same geography.

Task B1.3 — Rank: four burdens, four leaderboards

Most disasters, most deaths, most people affected and most money lost are four different maps

Top 12 countries on each measure, 1900–2022. The United States tops events and damage but ranks 23rd for deaths; Bangladesh is 3rd for deaths and 23rd for damage.

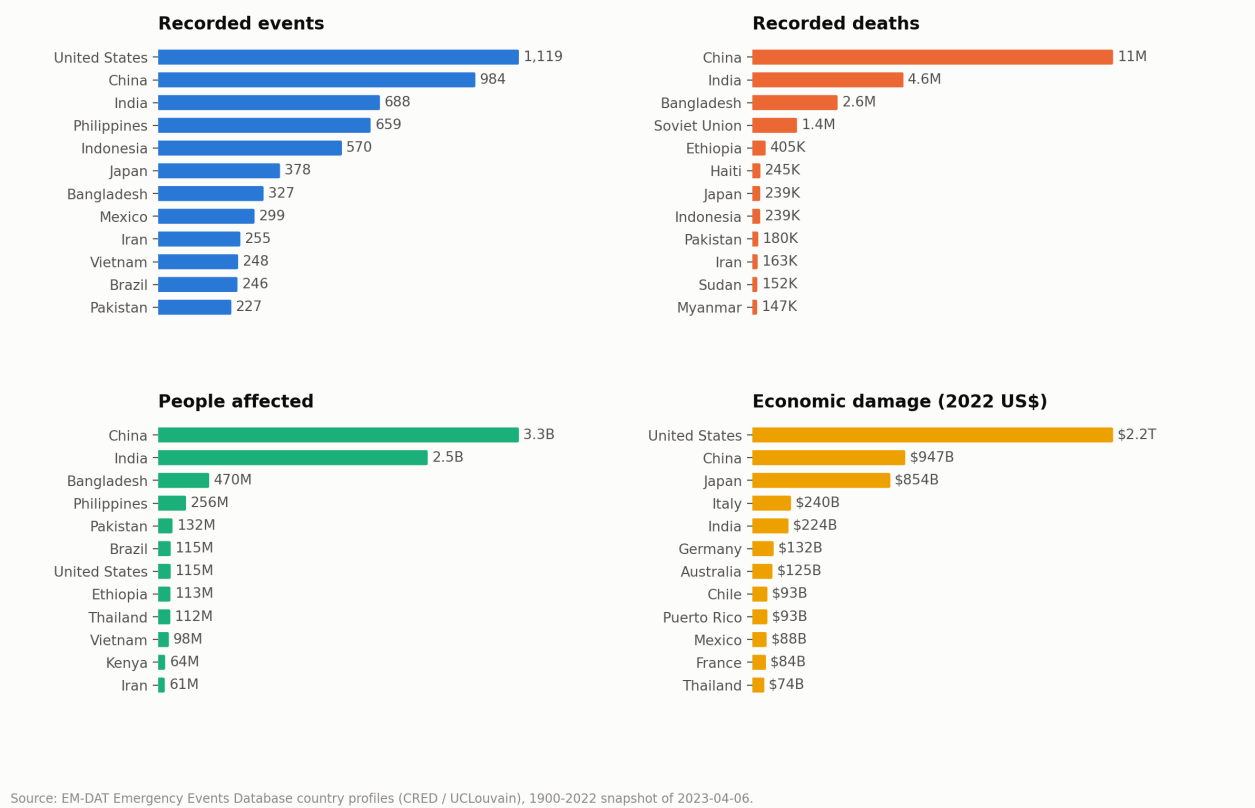


Figure 10. Top 12 countries on each of four measures. Small multiples of ranked bars, each panel on its own scale, because the question is rank *within* a measure rather than magnitude *across* measures. Every bar is directly labelled with its value, so no reading off an axis is required.

Inference. The four leaderboards barely overlap, and the mismatches are the finding:

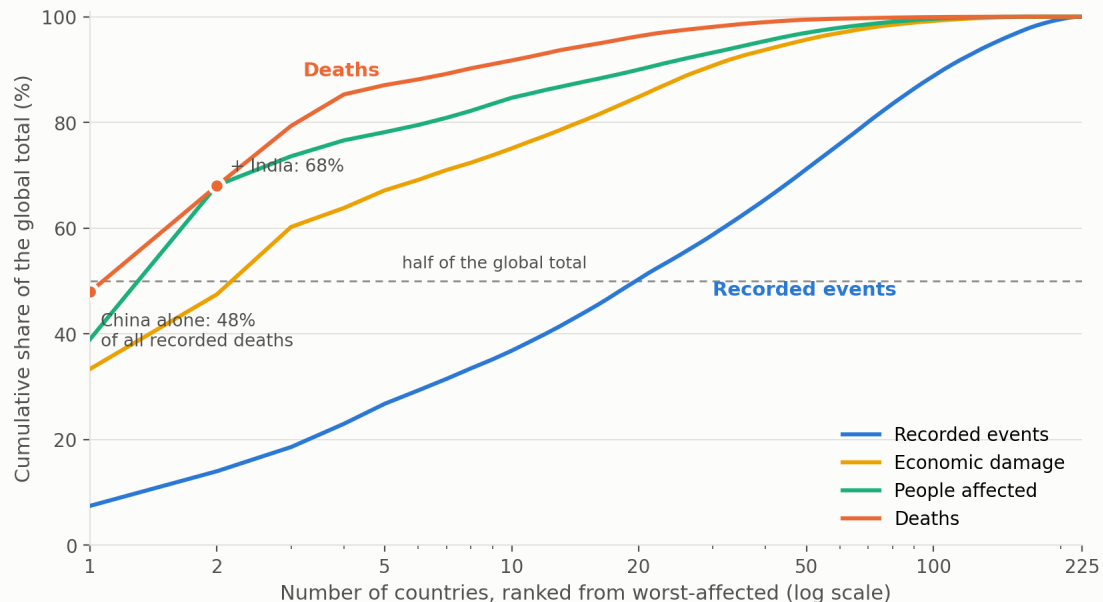
- The **United States** is 1st for events and 1st for damage (\$2.23 trillion) but **23rd** for deaths.
- **Bangladesh** is 3rd for deaths (2.59 million) but **23rd** for damage.
- **Australia** is 14th for events and **7th** for damage but **69th** for deaths.
- **Japan** is 3rd for damage (\$854 billion) but 31st for people affected.

Wealth appears to convert disaster exposure into property loss rather than into death. The Soviet Union appears 4th for deaths — a reminder that the country field contains historical states (Section 2.3, rule 7).

Task B1.4 — Derive: how concentrated is each burden?

Half of all deaths fall on 2 countries; half of all disasters are spread over 20

Cumulative share of each global total, countries ranked worst-first. Deaths and people affected are the most concentrated burdens; the raw count of events is by far the most evenly spread.



Source: EM-DAT Emergency Events Database country profiles (CRED / UCLouvain), 1900-2022 snapshot of 2023-04-06.

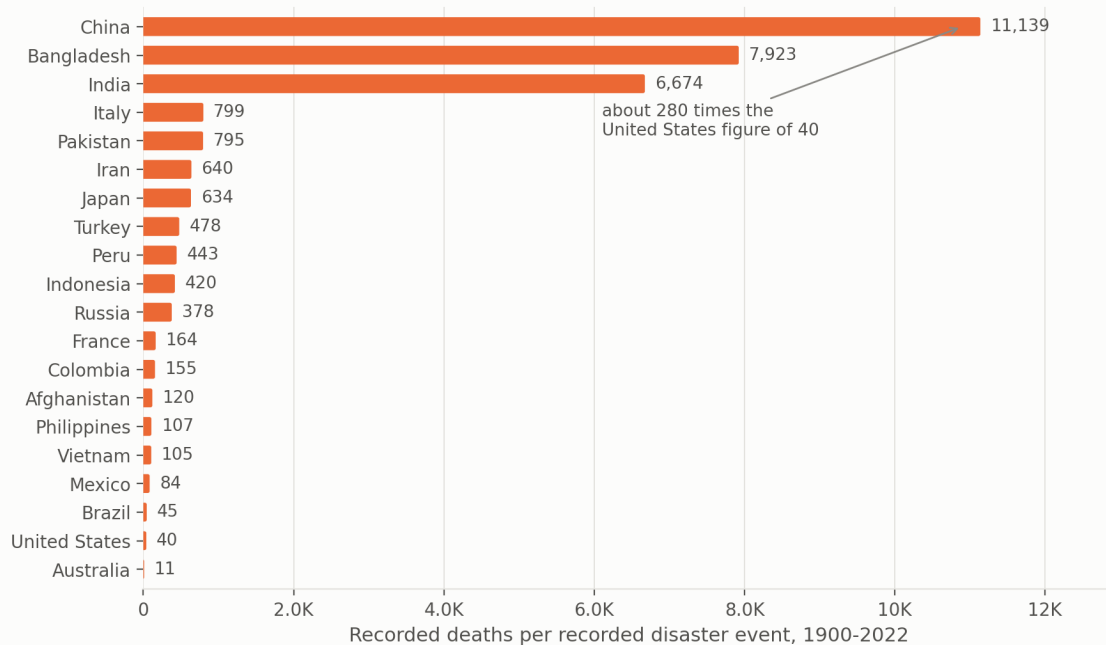
Figure 11. Cumulative share of each global total against the number of countries, ranked worst-first, on a logarithmic x -axis. This is a concentration (Lorenz-style) curve: the faster a curve rises, the fewer countries carry that burden.

Inference. The burdens are concentrated to very different degrees. It takes just **2 countries** to pass half of all recorded deaths (China alone is 48%), but **20 countries** to pass half of all recorded *events*. The top five countries hold **87.1%** of deaths, **78.2%** of people affected and **67.2%** of damage — but only **26.8%** of events. Disasters happen nearly everywhere; dying from them does not.

Task B1.5 — Derive: lethality among comparably exposed countries

Among equally disaster-prone countries, lethality differs by a factor of 1,000

The 20 countries with the most recorded events, ranked by deaths per event. Exposure is similar; the outcome is not - which points at vulnerability and response capacity, not at the hazards themselves.



Source: EM-DAT Emergency Events Database country profiles (CRED / UCLouvain), 1900-2022 snapshot of 2023-04-06.

Figure 12. Deaths per recorded event for the 20 countries with the most recorded events. Restricting to the most disaster-prone countries holds exposure roughly constant, so that the remaining variation reflects vulnerability rather than hazard frequency.

Inference. Among countries with broadly comparable exposure, lethality varies by a factor of about **1,000** — from China at 11,139 deaths per event and Bangladesh at 7,923, down to the United States at 40 and Australia at 11. China's figure is roughly **280 times** the United States'. Two caveats keep this honest: the file has no population denominator, so some of the China/India gap is simply population; and China's figure is heavily loaded by the pre-1960 mega-famines identified in Figure 7. Even after allowing for both, the spread is far too wide to be explained by hazard exposure, and points instead at building stock, early warning, and response capacity.

6 Task Set C — *What?* The Impact Signature of Hazards

Owner: Kalpit Phogat · Figures: 13–17 · Script: code/task_c_impact.py

Task C1.1 — Compare: each hazard's share of four burdens

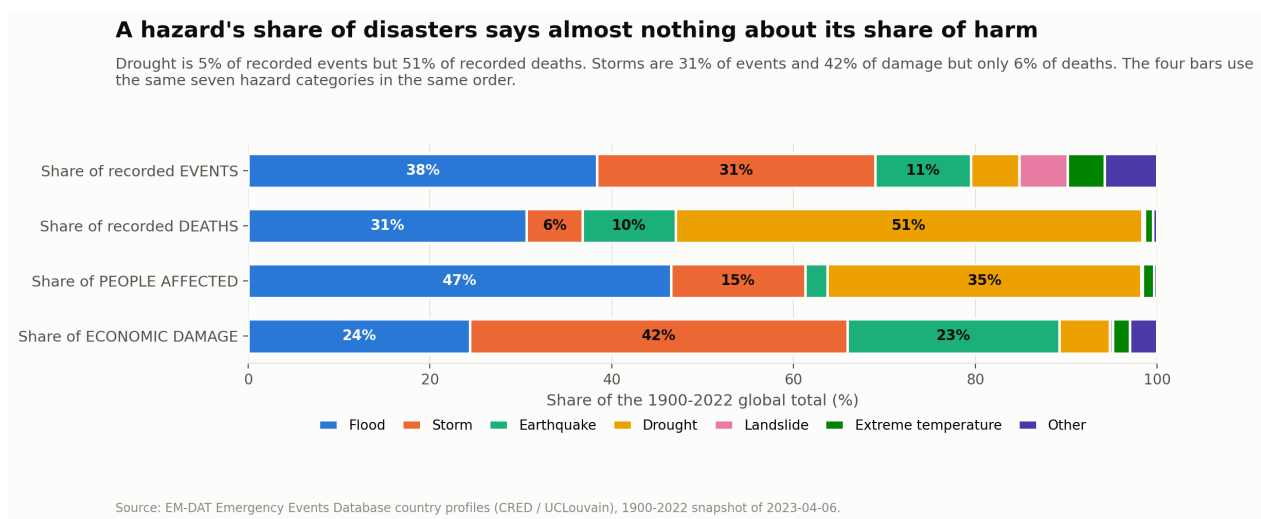


Figure 13. Share of the global total held by each hazard type, for four different burdens. 100%-stacked bars are used because the question is composition, and the four bars use the same seven categories in the same colour order so the eye can track a single hazard down the chart.

Inference. This is the key chart of Task Set C, and it shows that a hazard's share of *events* predicts almost nothing about its share of *harm* (Table 5). **Drought is 5% of events and 51% of deaths.** Storms are the mirror image: 31% of events and 42% of the money, but only 6% of the deaths. Earthquakes are the concentrated destroyer — a tenth of events, a quarter of the damage, but only 2.4% of people affected, because they strike hard over small areas. Floods are the great displacer: they affect the most people (46.5%) while being neither the deadliest nor the costliest per event.

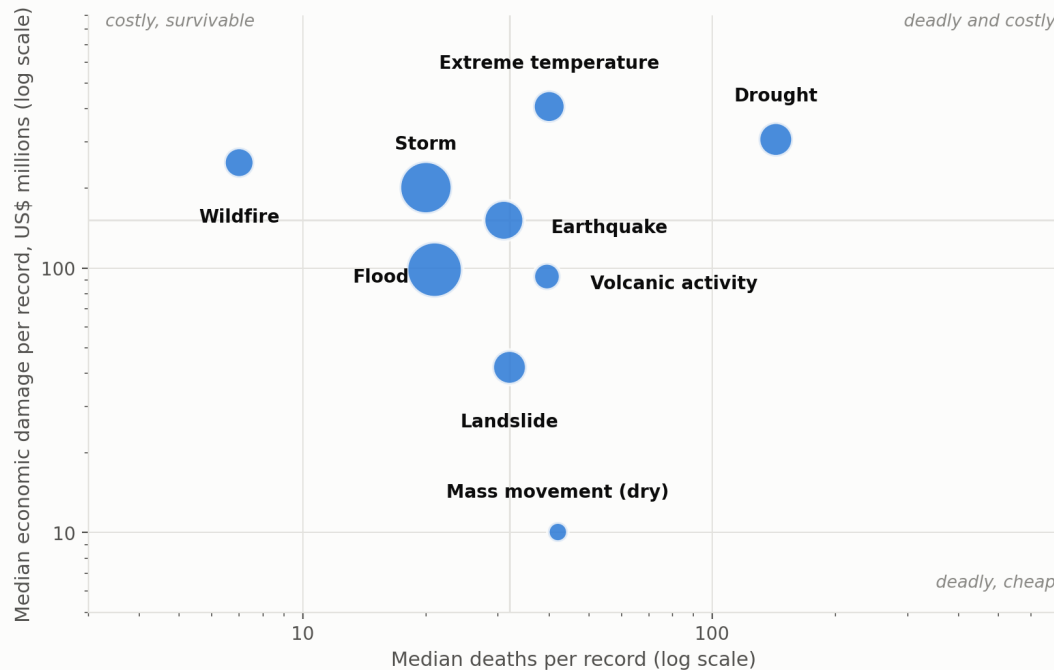
Table 5. Each hazard's share of four different global burdens, 1900–2022. The four columns are the four bars of Figure 13.

Hazard	Share of events	Share of deaths	Share of affected	Share of damage
Flood	38.4%	30.6%	46.5%	24.4%
Storm	30.6%	6.1%	14.8%	41.5%
Earthquake	10.5%	10.3%	2.4%	23.3%
Drought	5.3%	51.3%	34.5%	5.6%

Task C1.2 — Relate: deadly versus costly

Drought kills on a different scale from every other hazard

Each bubble is one hazard type; bubble size is the number of recorded events. A typical drought record carries 143 deaths, more than three times the next hazard - while extreme temperature carries the largest typical damage bill (\$408M). Medians, not totals, so one catastrophe cannot move a point.



Source: EM-DAT Emergency Events Database country profiles (CRED / UCLouvain), 1900-2022 snapshot of 2023-04-06.

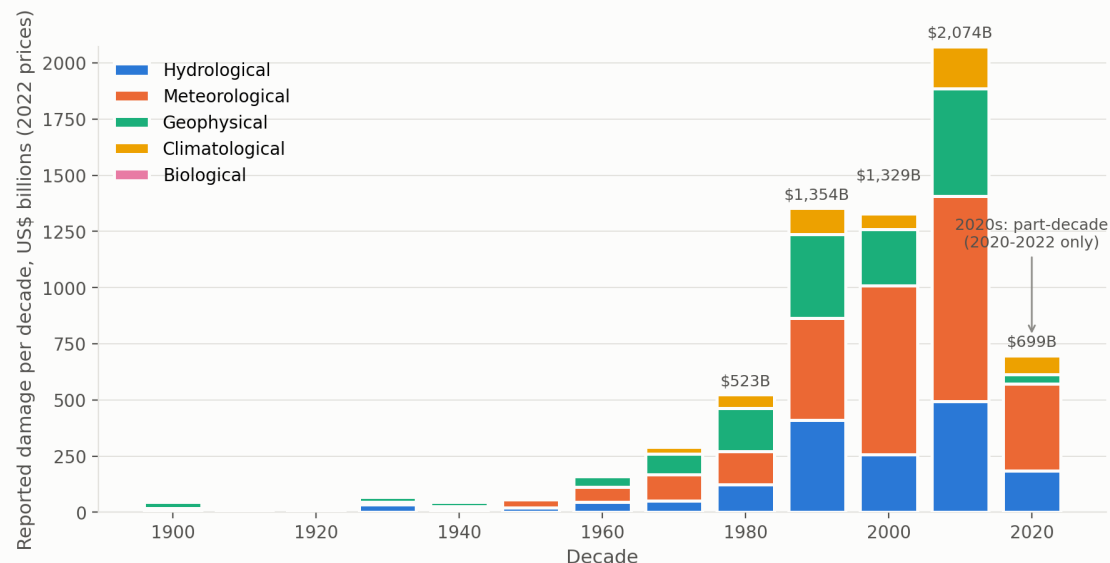
Figure 14. Median deaths against median damage per record, one bubble per hazard type, bubble area proportional to the number of events, both axes logarithmic. Medians rather than totals, so that a single catastrophe cannot move a hazard's position. A single hue is used because bubble identity is carried by direct labels, not by colour.

Inference. Drought sits alone on the right: a typical drought record carries **143 deaths**, more than three times the next hazard. The costliest *typical* hazard is a different one — extreme temperature, at a median **\$408 million** per record — and wildfire sits in the opposite corner from drought, expensive (a median \$250 million) but almost harmless to life (a median of 7 deaths). Hazards do not lie on a single “severity” axis; they occupy distinct positions in a deadliness–cost plane.

Task C1.3 — Trend: the real cost through time

The money cost of disasters keeps climbing, in real terms

Reported damage is inflation-adjusted to 2022 US dollars. Only 37% of records carry a damage figure, so these bars are lower bounds. Coverage does not improve after 1970 (it stays in a 32-48% band), so the rise cannot be an artefact of better reporting.



Source: EM-DAT Emergency Events Database country profiles (CRED / UCLouvain), 1900-2022 snapshot of 2023-04-06.

Figure 15. Reported economic damage per decade in constant 2022 US dollars, stacked by sub-group. Inflation adjustment is essential over a 123-year window; the sub-group breakdown shows which hazards drive the growth.

Inference. Real damage rises relentlessly: **\$1,354 billion in the 1990s, \$1,329 billion in the 2000s, and \$2,074 billion in the 2010s** — the costliest decade on record — and the 2020s are already at \$699 billion with only three years counted. Meteorological damage (storms) is the largest and fastest-growing component. Because the share of records carrying a damage figure does **not** improve after 1970, this rise cannot be dismissed as better reporting; if anything, flat-to-declining coverage means the true rise is steeper.

Task C1.4 — Compare: which hazards became survivable?

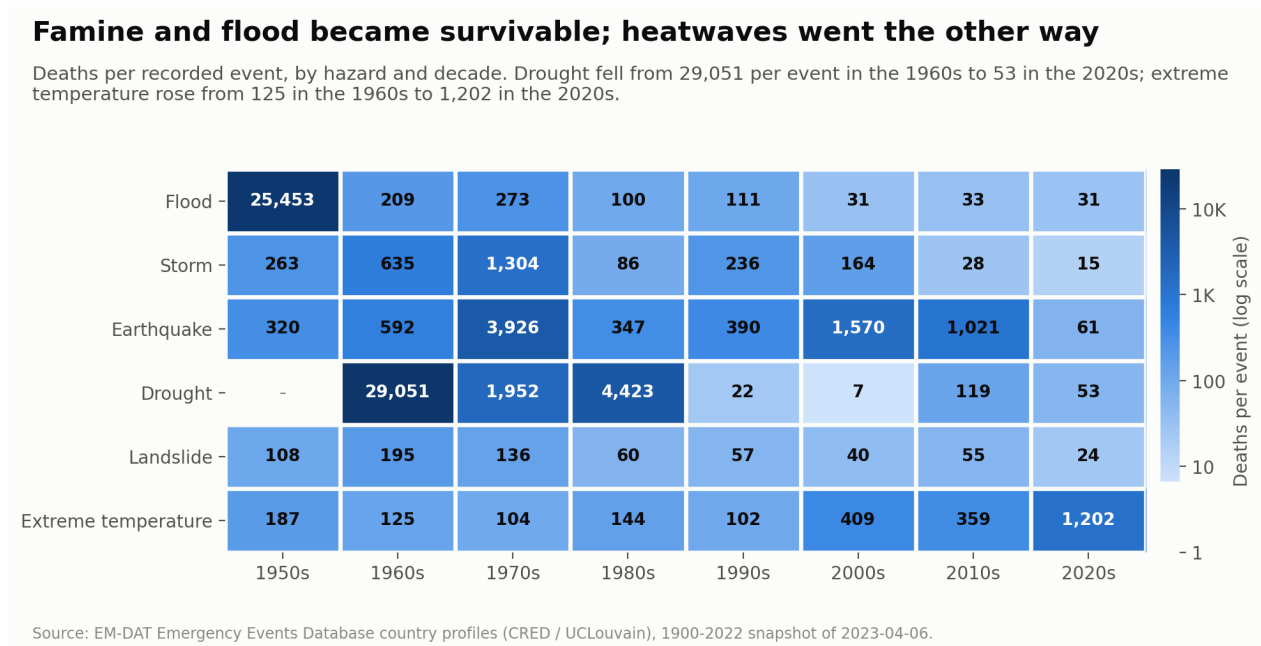


Figure 16. Deaths per recorded event, by hazard and decade, 1950s–2020s. A heatmap is the right form for a categorical × ordinal grid of a single measure; values are printed in every cell so the reader never has to estimate a number from colour, and the sequential blue ramp is logarithmic. The 2020s column covers 2020–2022 only.

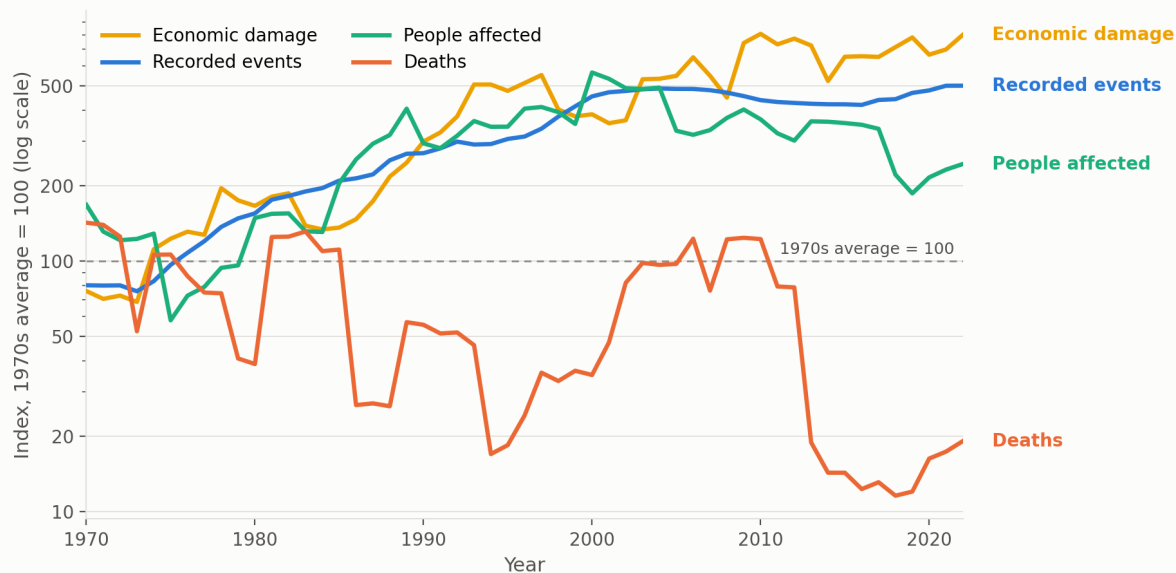
Inference. The improvement is real but uneven, and one hazard moves the wrong way:

- **Drought collapsed:** 29,051 deaths per event in the 1960s → 119 in the 2010s → 53 in the 2020s. Famine early-warning systems and food logistics are the plausible explanation.
- **Flood collapsed:** 25,453 in the 1950s → 33 in the 2010s.
- **Storm collapsed:** 1,304 in the 1970s → 28 in the 2010s → 15 in the 2020s. Cyclone forecasting and evacuation.
- **Earthquake did not improve:** 320 in the 1950s, 1,570 in the 2000s, 1,021 in the 2010s. Earthquakes give no warning, so the interventions that tamed floods and droughts do not apply.
- **Extreme temperature got worse:** 125 in the 1960s → 409 in the 2000s → **1,202 in the 2020s**. Heatwaves are the one hazard in this dataset becoming *more* lethal per event.

Task C1.5 — Correlate: the decoupling

Costs rose, deaths did not follow

Five-year moving averages, each indexed to its own 1970s mean so four different units share one axis. Damage grew 1.58x per decade ($p < 0.001$); annual deaths show no significant trend ($p = 0.064$), and the rank correlation between the two is -0.01 .



Source: EM-DAT Emergency Events Database country profiles (CRED / UCLouvain), 1900-2022 snapshot of 2023-04-06.

Figure 17. Recorded events, people affected, deaths and economic damage, 1970–2022, each as a five-year moving average indexed to its own 1970s mean. Indexing to a common base of 100 is what allows four different units to share a single axis honestly, instead of resorting to a dual axis.

Inference. Over 1970–2022 the four measures separate cleanly. Fitting a log-linear trend to each annual series gives Table 6. Events, damage and people affected all rise significantly. **Annual deaths show no statistically significant trend at all** ($p = 0.064$), and the Spearman rank correlation between annual deaths and annual damage is -0.01 — effectively zero. Median annual damage rose from \$36 billion (1970–89) to \$179 billion (2003–22), a five-fold increase, while median annual deaths went from 25,586 to 19,265. What *is* significant is deaths **per event**, falling by a factor of 0.58 each decade. The world is not suffering fewer disasters or cheaper ones; it is surviving each one far better.

Table 6. Log-linear trend fits to the annual series, 1970–2022. Exact p -values are reproduced in `code/facts_output.txt`.

Measure	Change per decade	p	R^2
Recorded events	$\times 1.43$	< 0.001	0.82
Economic damage (real)	$\times 1.58$	< 0.001	0.63
People affected	$\times 1.36$	< 0.001	0.29
Deaths	$\times 0.83$ — not significant	0.064	0.07
Deaths <i>per event</i>	$\times 0.58$	< 0.001	0.38

7 Supplementary Views: One Drill-Down per Task Set

Each task set above closes on a finding that invites one further question. This section answers those three questions with one additional figure each, in the same design conventions as Section 3. They are placed together here, after the three task sets, so that the figure numbering of Figures 3–17 — and every cross-reference to them in the text — stays exactly as it was.

Task A1.6 — Task Set A drill-down: which flood is actually rising?

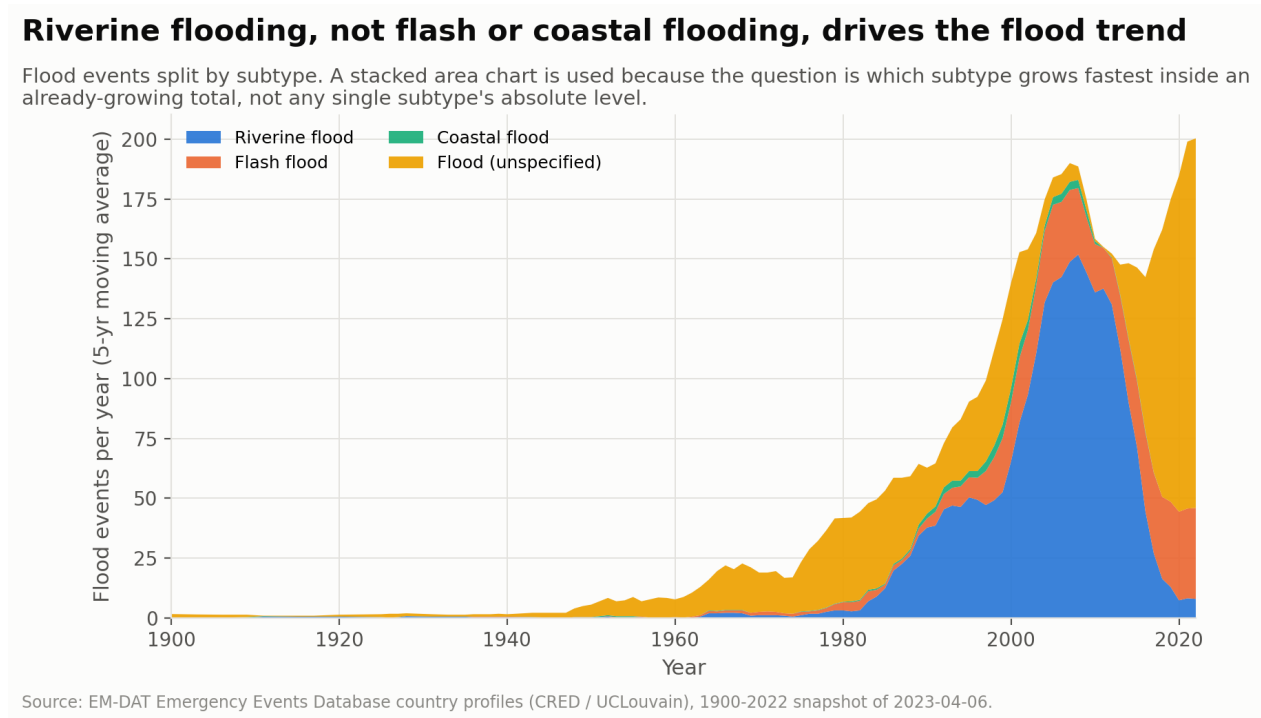


Figure 18. Flood events per year, split by subtype (5-year moving average). Figure 4 showed that floods drive most of the post-1970 rise; a stacked area chart is used again here, now one level down, because the question is which flood *mechanism* grows fastest inside that already-growing total, and stacking keeps the composition and the total legible in one image.

Inference. What is happening in this chart: the blue band (riverine flooding) is thin and flat until about 1970, then grows faster than every other band and becomes the largest single contributor to the whole stack by the 1990s. Of the 5,764 recorded flood events, riverine flooding accounts for **46.1%**, unspecified floods for 38.3%, flash floods for 14.2%, and coastal floods for only 1.5%. The practical reading is that the flood trend in Figure 4 is not a generic rise in “flooding” — it is overwhelmingly a rise in **river-basin flooding**, the subtype most directly linked to upstream land-use change, floodplain urbanisation, and extreme rainfall, which points mitigation spending toward watershed and river-basin management specifically rather than flood defence in general.

Task B1.6 — Task Set B drill-down: is exposure narrow or broad?

High event volume and broad hazard diversity are different things

The 20 countries with the most recorded events: total events (log x-axis) against how many distinct hazard types each has ever recorded. A scatter plot is used because the question is the joint distribution of two independent measures, not a ranking on either alone.



Source: EM-DAT Emergency Events Database country profiles (CRED / UCLouvain), 1900-2022 snapshot of 2023-04-06.

Figure 19. Total recorded events (log x -axis) against the number of distinct hazard types ever recorded, for the 20 most-exposed countries, coloured by each country's single most frequent hazard. A scatter plot is used, not a ranked bar chart, because the question is the joint distribution of two independent measures — volume and diversity — and a bar chart can only rank one measure at a time.

Inference. What is happening in this chart: the points do not fall on a rising diagonal, which is the finding itself — volume and diversity are not the same axis. **India (688 events) and Peru (195 events) sit highest on the diversity axis with 10 distinct hazard types each,** even though India ranks only 3rd by volume and Peru 16th. The two highest-volume countries, the United States (1,119 events) and China (984), sit lower, at 8 and 9 types respectively. India and Peru owe their diversity to spanning several climate and geological zones at once (Himalayan, coastal and arid for India; coastal, Andean and Amazonian for Peru), whereas the United States and China's totals are driven by fewer, very frequently repeated hazards (storms and floods). The practical implication is that a single national risk score conflates two different planning problems: a high-volume, low-diversity country needs deep capacity against one or two hazard types, while a high-diversity country needs broad capacity across many.

Task C1.6 — Task Set C drill-down: what does the CPI adjustment actually do?

Why every damage figure in this report is inflation-adjusted

Left: the CPI series used for adjustment, shown directly rather than only through its effect. Right: the same raw damage series before and after adjustment - the nominal series understates old disasters relative to recent ones by exactly the ratio on the left.

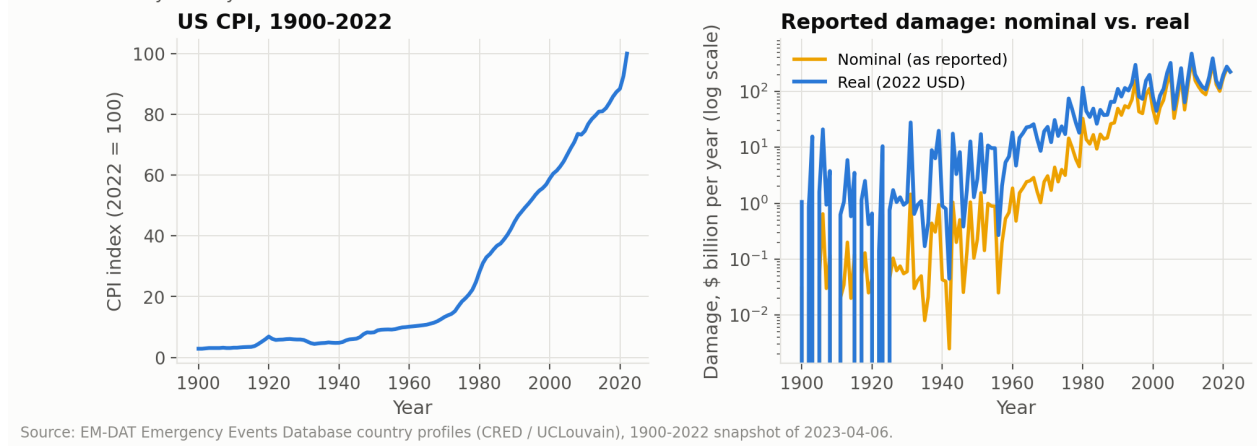


Figure 20. Left: the US CPI series used throughout this report for inflation adjustment, shown directly. Right: the same raw annual damage total before (nominal) and after (real, 2022 USD) adjustment. Every other damage figure in this report uses only the adjusted series; this is the one figure that shows the adjustment mechanism itself, because a reader should be able to see what the correction does, not just trust that it was applied.

Inference. What is happening in this chart: the left panel is a smooth, monotonically rising curve, which is expected of a price index; the right panel's two lines diverge visibly as the years get older, with the real (blue) line running consistently above the nominal (orange) line before 1980. The CPI itself rose **35-fold** from 1900 (index 2.8, base 2022 = 100) to 2022 (index 100), so a disaster reported as causing \$1 billion of damage in 1900 represents roughly \$35 billion of real 2022 purchasing power — the gap between the two lines in the right panel at any given year is exactly that year's CPI ratio to 2022, no more and no less. This is the reason every damage-based claim elsewhere in this report (Task Set C, Figures 15 and 17, and the Task Set B leaderboards) is stated in real terms: a raw, unadjusted comparison across a 123-year window would silently understate every disaster before roughly 1980 relative to one after it.

8 The Data Story

Read in sequence, the fifteen charts tell one story in four movements.

1. The record grows — but so does the recording. Figures 3–5 establish what the dataset contains and immediately undercut the naive reading of it. Recorded events grew 52-fold from the 1900s to the 2000s, but 91% of the record falls in 43% of the years, and damage reporting covers at most half the records. Counts are unreliable; rates are not.

2. Yet the world got dramatically better at surviving. Figure 6 shows deaths per recorded event falling 99.4% across the century, and Figure 17 confirms it is the only mortality trend in the modern era that is statistically significant. Figure 16 localises the improvement: droughts, floods and storms — the hazards that give warning — became survivable, while earthquakes, which do not, did not improve at all.

3. But “the world” is the wrong unit. Figures 8–12 show the improvement is not evenly distributed. Occurrence is global; death is not. Two countries carry half of all recorded deaths, and

among the twenty most disaster-exposed countries lethality spans three orders of magnitude. The United States leads the world in recorded disasters and in dollar losses while ranking 23rd in deaths — wealth converts disaster exposure into property damage instead of mortality.

4. And the bill keeps growing. Figures 13–15 show that the hazards that kill (drought) and the hazards that cost (storms, earthquakes) are largely different hazards, and that real damage reached a record \$2.07 trillion in the 2010s. Figure 17 closes the loop: costs and deaths have decoupled almost perfectly (rank correlation -0.01).

Answer to the guiding question. The burden of natural disasters has grown, but not in the way the headline suggests. *Exposure* has risen — more assets, more people, and much better reporting. *Vulnerability* has fallen sharply, and that fall is the real achievement of the last century. And the fall is unevenly shared, incomplete for hazards that give no warning, and reversing outright for extreme heat.

And one honest caveat on our own headline, from Figure 7: most of the century-scale fall in *total* deaths is the disappearance of a handful of mega-catastrophes rather than a broad decline in ordinary disaster deaths. The per-event improvement is robust; the total-deaths improvement is more fragile than it first appears.

9 Consolidated Inferences

1. Floods and storms are 69% of all recorded events; drought is 5.3% of events but 51.3% of deaths.
2. Recorded events grew $52\times$ from the 1900s to the 2000s, but 91% of the record sits in 43% of the years — the count measures reporting as much as hazard.
3. Deaths per recorded event fell 99.4%, from $\approx 19,900$ (1900s) to ≈ 129 (2010s).
4. That fall is driven by the disappearance of mega-catastrophes; excluding each decade's three worst records, residual deaths show no downward trend (2000s: 390k > 1900s: 117k).
5. All 225 countries appear in the record, but 2 countries hold half of all deaths and 5 countries hold 87.1%.
6. The United States is 1st in events and damage but 23rd in deaths; Bangladesh is 3rd in deaths but 23rd in damage.
7. Among the 20 most disaster-exposed countries, deaths per event span a factor of $\approx 1,000$ (China 11,139 $\rightarrow$ Australia 11).
8. A hazard's share of events does not predict its share of harm: storms are 31% of events but 6% of deaths; drought is 5% of events but 51% of deaths.
9. Real economic damage reached a record \$2.07 trillion in the 2010s and is rising $\times 1.58$ per decade ($p < 0.001$).
10. Annual deaths show no significant trend since 1970 ($p = 0.064$); the rank correlation between annual deaths and annual damage is -0.01 . Costs and mortality have decoupled.
11. Drought lethality fell from 29,051 to 53 deaths per event; earthquake lethality did not improve; **extreme-temperature lethality rose roughly ten-fold** (125 $\rightarrow$ 1,202).
12. Wealth appears to convert disaster exposure into property loss rather than death — the clearest policy-relevant pattern in the dataset.

10 Limitations and Threats to Validity

- **Reporting bias is not correctable.** We mitigate it by preferring rates to counts and by restricting statistical trend fits to 1970 onward, but pre-1970 counts remain unusable as a hazard measure.
- **Damage is under-reported** (36.9% coverage), so all damage totals are lower bounds. Coverage does not improve after 1970, so the upward trend is not a coverage artefact.
- **No population denominator.** Deaths per event is a severity proxy, not a risk rate. China and India’s dominance in Figure 9 partly reflects population, and this is stated wherever the figure is discussed.
- **Rows are aggregates**, so per-event values are cell averages, not per-disaster values.
- **Historical states** (Soviet Union, Yugoslavia, Czechoslovakia and four others) appear as countries and are flagged in the cleaned extract.
- **The 2020s are a part-decade** (2020–2022) and are annotated as such in Figures 5, 6, 15 and 16.
- **No epidemics.** COVID-19 and all disease events are absent from this file; nothing here speaks to biological disaster risk.
- **Correlation is not causation.** Section 8’s attribution of falling lethality to warning systems and response capacity is an interpretation consistent with the pattern in Figure 16, not something this dataset can establish on its own.

11 Tools and Reproducibility

Table 7 lists the tool used for each component of the submission.

Table 7. Tools used for each component of the submission.

Component	Tool
Data cleaning and analysis	Python 3.12, pandas 3.0, NumPy, SciPy
Statistical charts (Figures 3–7, 10–17)	Matplotlib 3.11
Method diagrams (Figures 1, 2)	Matplotlib 3.11
Choropleth maps (Figures 8, 9)	Plotly 6.9 with Kaleido static export
Interactive dashboards and video demo	Tableau Public, reading <code>data/emdat_clean_tableau.csv</code>
This report	L ^A T _E X (pdf _l atex), source <code>report/DAS732_A1_Report.tex</code>

To regenerate every figure and rebuild this PDF from the raw CSV:

```
cd code
python run_all.py
```

`code/facts.py` and `code/facts2.py` re-derive **every quantitative claim** made in this report; their output is saved as `code/facts_output.txt` and `code/facts2_output.txt` so that any number here can be traced to the line that produced it. `code/check_report.py` then re-reads this L^AT_EX source and asserts every figure quoted in it against the data, so a stale number cannot survive a rebuild. `Tableau_Guide.md` gives step-by-step instructions for rebuilding six demonstrable views in Tableau from the cleaned extract, for the video demo.

12 Author Contributions

All three members contributed equally. Task sets were assigned as whole units — a member owns their sub-question, their figures, their inferences and their section of this report and of the video — so that no task is split across members.

Lakshya Jain (BT2024044) — Data processing lead and Task Set A (*When?*)

Sourced the dataset and identified the delimiter/decimal and trailing-whitespace issues; wrote the shared cleaning module `emdat_common.py` including the 2023 part-year exclusion and the CPI-adjustment verification; produced the Tableau-ready extract and the pipeline diagram (Figure 2); designed and implemented Task Set A (Figures 3–7), including the reporting-bias diagnostic (Figure 5) and the trimmed-deaths robustness test (Figure 7) that qualified the team’s headline finding, and the flood-subtype drill-down (Figure 18); wrote Sections 2, 3 and 4; presents the preprocessing minute and Task Set A in the video.

Arnav Jain (BT2024233) — Task Set B (*Where?*)

Designed and implemented Task Set B (Figures 8–12); built the Plotly choropleth pipeline including the logarithmic colour scale and ISO-3 joins; produced the four-panel country leaderboards, the concentration curves and the lethality ranking and the volume-versus-diversity scatter (Figure 19); identified the events/deaths/damage rank mismatch (United States 1st versus 23rd) that became a central thread of the data story; wrote Sections 5 and 9; presents Task Set B in the video.

Kalpita Phogat (BT2024093) — Task Set C (*What?*)

Designed and implemented Task Set C (Figures 13–17); built the hazard impact-signature chart, the deadliness–cost plane, the real-damage time series, the lethality heatmap and the indexed decoupling chart; ran the log-linear trend fits and the Spearman decoupling test, and established that annual deaths carry no significant post-1970 trend; found the rising extreme-temperature lethality; produced the CPI-adjustment explainer (Figure 20); wrote Sections 6 and 8; presents Task Set C in the video.

Jointly

The guiding question and the three-way task decomposition (Figure 1); the visual design conventions of Section 3; Sections 1, 7 and 10; the L^AT_EX report template; the video storyboard; final review.

References

- [1] H.-J. Schulz, T. Nocke, M. Heitzler and H. Schumann, “A design space of visualization tasks,” *IEEE Transactions on Visualization and Computer Graphics*, vol. 19, no. 12, pp. 2366–2375, 2013.
- [2] Centre for Research on the Epidemiology of Disasters (CRED), UCLouvain, *EM-DAT: The International Disaster Database*. <https://www.emdat.be>
- [3] Kaggle dataset mirror, *Natural Disasters Emergency Events Database*. <https://www.kaggle.com/datasets/mexwell/natural-disasters-emergency-events-database>